Case B

Mr. O'Shea — 72-Year-Old Retired Librarian | Working Diagnosis: Osteoporotic Vertebral Compression Fracture (T7–T8)

Musculoskeletal Physiotherapy

■ SUBJECTIVE PROBLEM LIST

- Physical Impairments
- Mid-thoracic pain VAS 4/10 — worse with prolonged standing/loading
- Tenderness and percussion pain at T7–T8 (fracture site)
- Height loss ("shrinking") — loss of vertebral body height
- Likely reduced thoracic and hip extension ROM
- Likely reduced back extensor and core muscle strength
- Difficulty with activities requiring thoracic loading (lifting, carrying)
- Possible difficulty with dressing (bending forward)
- Reduced walking tolerance and exercise capacity
- Potential social isolation secondary to pain and reduced mobility

■■ RED FLAGS

- Clinical Significance
- Action if Present
- Bilateral lower limb weakness, paraesthesia, gait disturbance, or bladder/bowel dysfunction
- Spinal cord compression (myelopathy) or cauda equina syndrome from fracture fragment migration. EMERGENCY — can cause permanent paralysis.
- DO NOT MOBILISE. Emergency referral to hospital (999/112). Urgent MRI.
- Severe, constant, unremitting back pain — worse lying down, disturbing sleep
- Malignancy (primary or metastatic), particularly in 72M. Myeloma, prostate, lung, and renal metastases are common thoracic spinal tumours.
- Urgent GP referral for bloods (PSA, ESR, serum protein electrophoresis, full blood count), plain X-ray, and possible MRI.

■ PATHOPHYSIOLOGY (KEY POINTS)

- ▼ Structural & Biomechanical Mechanism Osteoporosis is characterised by reduced bone mineral density (BMD) and disrupted trabecular microarchitecture, resulting in decreased bone strength and increased fracture susceptibility.
- The thoracic vertebrae (T7–T8) are vulnerable sites for osteoporotic VCF because the natural thoracic kyphosis directs axial compressive load predominantly onto the anterior vertebral body — the region with the least cortical bone support.
- Localised periosteal nerve endings and the sinuvertebral nerve at the fracture site generate deep, aching nociceptive pain, which is exacerbated by axial loading (standing, extension) and relieved by offloading (lying down).
- Mid-thoracic pain worsening with prolonged standing: Upright posture loads the anterior wedged vertebral body — fracture site becomes compressed, stimulating periosteal nociceptors.
- Pain on spinous process percussion: Vibration transmitted to the fractured vertebra triggers periosteal nerve activation at the fracture site — highly specific for bony pathology.
- Progressive kyphosis / "shrinking": Anterior wedge collapse reduces vertebral body height — cumulative across one or more levels results in measurable height loss and visible kyphotic deformity.

■ DIFFERENTIAL DIAGNOSES

See full case notes

■ OBJECTIVE ASSESSMENTS & HOW TO PERFORM

Key Observation Findings

Posture: Increased thoracic kyphosis (flexed posture of thoracic spine) — visible as a "hump" or "dowager's hump." Compensatory forward head posture and increased lumbar lordosis are common secondary adaptations. Reduced lumbar extension range. Movement: Thoracic extension will be restricted and painful (loads the fracture site). Thoracic flexion may be less provocative but also limited. Lateral flexion and rotation may be preserved but guarded. Observe for paraspinal muscle guarding/splinting bilaterally at the T7–T8 level. Gait:

TEST 3

Lower Limb Neurological Screen (Reflexes, Myotomes, Dermatomes) + Babinski Sign What it Measures Thoracic spinal cord integrity — screens for cord compression (myelopathy) from posterior VCF wall retropulsion or thoracic canal stenosis. How to Perform 1. Lower limb reflexes: Patellar (L3–L4), Achilles (S1–S2) — compare bilaterally with reflex hammer. 2. Myotome screen: Hip flexion (L1–L2), knee extension (L3), dorsiflexion (L4), great toe extension (L5), plantarflexion (S1). 3. Dermatome light touch: medial calf (L4), dorsum foot (L5), lateral foot (S1). 4. Babinski sign: Stroke lateral plantar surface of foot from heel to MTP joints. Upward (extensor) great toe response = positive UMN sign.

■ TREATMENT PLAN

- 2. Did you have a specific incident or injury, or did the pain come on gradually? Have you had a fall recently?
- 3. What movements or positions aggravate and relieve your pain?
- 4. You mentioned feeling like you are "shrinking" and your posture is changing — when did this start, and has it been gradual or sudden?
- 5. Do you have any weakness, numbness, or changes in your bladder or bowel function?
- 6. How long have you been on corticosteroids, what dose, and are you on any bone protection medication (e.g., bisphosphonates)?
- Absent or decreasing reflexes in lower limbs + upper motor neuron signs (clonus, Babinski)

• Thoracic cord involvement — thoracic myelopathy. The thoracic cord has a narrow spinal canal; even small fracture fragments can compress the cord.
• Emergency neurosurgical referral. Do NOT mobilise or exercise the thoracic spine.
• Key Distinguishing Clinical Features
• VCF vs. Malignancy:
• VCF vs. Thoracic Disc Pathology:
• Corticosteroid-induced osteoporosis vs. Primary (post-menopausal/senile):